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Project Proposal
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Graph-Based Spectral Analysis of Paddy Farming Practices

Problem Statement

Agricultural production depends on a complex combination of factors including soil conditions, fertilizer usage, weather variability, and cultivation practices. Farmers often apply different combinations of farming techniques (such as fertilizer combinations, seed rates, and different crop varieties) under varying environmental conditions. However, these farming strategies are rarely analyzed systematically to understand whether similar farms exhibit comparable cultivation patterns.

This project proposes modeling farms as a similarity network, where each farm is represented as a node and edges represent similarity in farming practices and environmental conditions. By constructing this network and applying spectral clustering, the project aims to identify groups of farms that share similar farming strategies. The project will also explore anomaly detection to identify farms that deviate significantly from typical patterns, which may indicate unusual practices or environmental conditions.

Dataset

The dataset used in this project is the [Paddy Dataset] from the UC Irvine Machine Learning Repository, donated on July 14, 2025. The dataset contains 2,790 observations collected from paddy (rice) farms and includes 45 variables describing various aspects of cultivation practices and environmental conditions.

The variables include information related to:

- farm characteristics (e.g., hectares, agriblock)
- crop variety
- soil types
- seed rate and nursery practices
- fertilizer applications (e.g., DAP, urea, potash, micronutrients)
- pesticide usage
- drainage indicators
- weather variables such as minimum and maximum temperature
- wind conditions and other environmental measurement

These variables capture a comprehensive view of farming practices and environmental influences affecting rice production.

Proposed Methodology

Data Processing

All numeric variables will first be standardized to ensure that features measured on different scales (kg, tonnes, days) contribute equally to similarity calculations. Categorical variables (soil types, crop variety, agriblock) will be encoded using appropriate encoding techniques such as one-hot encoding. Missing values, if present, will be handled using standard imputation techniques to ensure that all observations can be used in the analysis.

Dimensionality Reduction using PCA

The dataset contains more than 45 variables but only 2,790 records. Given that one-hot encoding will introduce additional columns, the dataset may face **High-Dimension, Low-Sample Size (HDLSS)** challenges. In addition to basic feature selection using Lasso (L1 regularization) to remove irrelevant features, **Principal Component Analysis (PCA)** will be used to reduce the dimensionality of the dataset.

PCA transforms a large, correlated dataset into a smaller set of uncorrelated variables called **principal components**, maximizing variance retention while preserving important patterns in the data. This dimensionality reduction step helps improve computational efficiency and reduces noise before constructing the similarity graph.

Similarity Graph Construction, Spectral Clustering, and Anomaly Detection

A similarity graph will be constructed where each node represents a farm and edges represent similarity between farms. Similarity will be computed using a Gaussian similarity function based on Euclidean distance in the reduced feature space.

Once the graph is constructed, spectral clustering will be applied to identify clusters of farms. The graph Laplacian matrix, derived from the similarity matrix, will be used to compute eigenvectors that represent a low-dimensional embedding of the graph structure. These eigenvectors capture the underlying connectivity of the farm network.

After obtaining this spectral embedding, standard clustering algorithms such as k-means will be applied to identify clusters of farms. These clusters are expected to reveal distinct farming strategies or environmental conditions shared by groups of farms.

After clusters are identified, anomaly detection techniques such as distance-based outlier detection or graph connectivity measures will be applied. This step aims to identify farms that deviate significantly from typical patterns or cluster structures.

Planned Evaluation Strategies

Since the project primarily focuses on unsupervised learning, evaluation will emphasize the quality and interpretability of the discovered structures.

Cluster Validation Metrics

Clustering quality will be evaluated using internal clustering metrics such as within-cluster variance, which measures how similar farms are within the same cluster compared to farms in different clusters.

Baseline Comparisons

To evaluate the effectiveness of spectral clustering, results will be compared with standard clustering approaches such as k-means clustering and hierarchical clustering. These comparisons help determine whether the graph-based approach captures structures that traditional clustering methods may fail to identify.

Visualization and Interpretation

PCA will also be used to visualize clusters in two-dimensional space. Visualization can serve as a qualitative evaluation method to help interpret whether clusters correspond to meaningful farming strategies.

Feature distributions within each cluster will also be examined to understand how farming practices differ across clusters. Additionally, detected anomalies will be analyzed to determine whether they correspond to unusual farming practices or environmental conditions.